

AI ENGINEER ROADMAP

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Starting from Month 4

COMPLETED Month 1: Python depth Month 2: Data foundations Month 3: AI / LLM conceptual pass

Active Roadmap

MONTH 4 // RAG Systems

LLM + retrieval + production

- + OpenAI / Anthropic API integration
- + Embedding models: text-embedding-3-small, all-MiniLM
- + Qdrant: create collections, embed vectors, write queries
- + LangChain / LlamaIndex: build a full RAG pipeline
- + Prompt engineering: few-shot, chain-of-thought, system prompt
- + LLM output validation with Pydantic + structured output
- Memorizing every LangChain feature — use only what you need
- Pinecone paid plan — start with Qdrant local

PROJECT Embed Space Science papers into Qdrant -> RAG assistant -> serve via FastAPI

! This is your strongest portfolio project. Invest in documentation and a clean README.

MONTH 5 // Agent Systems & Automation

Turn an LLM into an autonomous system

- + LangGraph: stateful agent pipelines with memory
- + Tool calling / Function calling — give your LLM tools
- + Multi-agent orchestration basics
- + n8n: AI workflow automation
- + Guardrails: hallucination detection, output filtering
- Kafka / RabbitMQ — too early for AI Engineer stage
- Apache Spark / Databricks — Data Engineer territory

PROJECT Agent that analyzes Digital Twin sensor data and answers anomaly questions in natural language

MONTH
6

// MLOps, Security & Final Capstone

Ship a production-grade AI system

+

MLflow: experiment tracking and model versioning

+

GitHub Actions: CI/CD pipeline for ML

+

AI Security: Prompt Injection detection and defense

+

LLM Monitoring: latency, token usage, error rates

+

Docker Compose: spin up the full stack with one command

-

SageMaker / Vertex AI — black boxes, not needed yet

-

Kubernetes deep dive — DevOps territory, not AI Engineer

PROJECT

Orbital Guardian: PyTorch + Qdrant + RAG agent + FastAPI + Docker Compose — full stack

!

Prompt Injection and AI Security are rare in portfolios. This is where you stand out.

Portfolio Targets

#	PROJECT	STACK	SIGNAL
1	Space Science RAG Assistant Document Q&A system	Qdrant LangChain FastAPI Docker	RAG + production deployment
2	OCR -> TrOCR Upgrade Transformer-based modernization	HuggingFace TrOCR FastAPI pytest	Model integration + testing
3	Orbital Guardian AI Sensor anomaly + RAG agent	LangGraph MLflow Docker Compose n8n	Agent + MLOps + full stack

Rules

+

README + architecture diagram on every project

+

Measure latency, token count, error rate — always

+

2-3 deep GitHub repos > 10 shallow ones

+

Write Prompt Injection tests — rare differentiator

+

Submit a PR to LangChain or LlamaIndex

-

Collecting Coursera certificates — nobody cares

-

Chasing every new LLM tool — master the core first

-

Kaggle competitions — zero value for AI Engineer roles

-

Pushing tutorial projects to GitHub

-

Switching roadmaps before finishing a phase

